

Multiple Choice (10 marks)

1. What is the maximum number of phases that can be at equilibrium with each other in a three component mixture?
 - (a) 6
 - (b) 9
 - (c) 2
 - (d) 4
 - (e) 10
2. Calculate the de Broglie wavelength for (a) an electron with a kinetic energy of 100eV
 - (a) 0.175 nm
 - (b) 0.080 nm
 - (c) 0.098 nm
 - (d) 0.123 nm
 - (e) 0.135 nm
3. One cubic millimeter of oil is spread on the surface of water so that the oil film has an area of 1 square meter. What is the thickness of the oil film in angstrom units?
 - (a) 200 Å
 - (b) 1 Å
 - (c) 0.1 Å
 - (d) 50 Å
 - (e) 500 Å
4. Which of the following molecules is a strong electrolyte when dissolved in water?
 - (a) CH_3COOH
 - (b) CO_2
 - (c) O_2
 - (d) H_2O
 - (e) NH_3
5. Which of the following is an n-type semiconductor?
 - (a) Silicon
 - (b) Boron-doped silicon

- (c) Pure germanium
- (d) Arsenic-doped silicon
- (e) Indium Phosphide

True / False (10 marks)

Answer True or False

6. True or False: The nuclide ^{31}P has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field.
7. True or False: The temperature change of a diatomic gas can be directly calculated as 13.5 K from the given values of q and w .
8. True or False: When two coins are tossed, the expected number of heads, denoted as $\langle w \rangle$, is 1.
9. True or False: Element X has a valence of +4, indicating it can accept four electrons in reactions.
10. True or False: The carbon configuration $1s^2 2s^2 2p^2$ corresponds to a total of 20 microstates in quantum mechanics.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the kinetic energy of a particle with a mass of 1.0 g released from rest near the Earth's surface, where $g = 8.91 \text{ m/s}^2$, after 1.0 s. Provide a detailed analysis and calculations.
12. Discuss the total binding energy of the carbon-12 nucleus and analyze the significance of the average binding energy per nucleon in the context of nuclear stability.

End of Paper